

Categorical Data Analysis Assignment

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1 Basic EDA using Lung Disease Data

1.1 Py Code

```
# current working directory
import os
print(os.getcwd())
print()

# install necessary packages
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import scipy.stats as stats

# read data
df = pd.read_csv("lung_disease.csv")
df.head()
```

1.2 Output

ID	Smoking	Age	Income	Pollution	LungDisease
1	Yes	36	Low	High	Yes
2	No	28	Low	Low	No
3	No	52	Medium	High	No
4	No	39	High	High	No
5	Yes	32	Low	Low	Yes

2 Age Distribution — Normal or Skewed

2.1 Py Code

```
# Age distribution
print(df['Age'].describe())

# plot
sns.histplot(df['Age'], kde=True)
plt.title("Age Distribution")
plt.show()

# Skewness check
print("Skewness:", df['Age'].skew())
```

2.2 Output

```
count    500.000000
mean      43.904000
std       14.604841
```

```

min      20.000000
25%     30.750000
50%     45.000000
75%     56.000000
max      69.000000

```

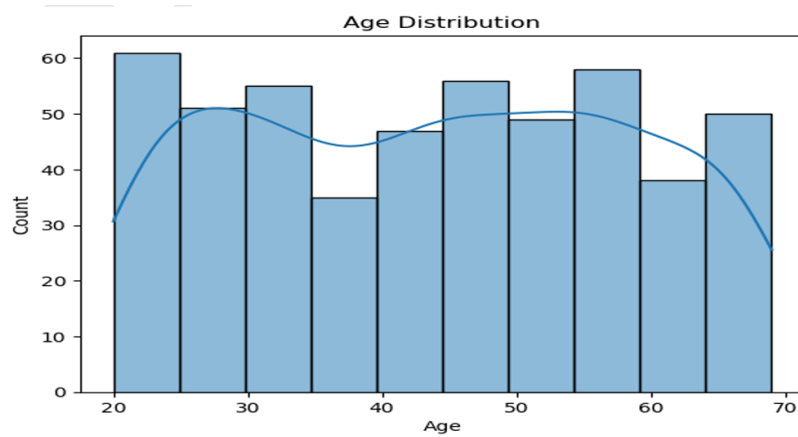


Figure 1: Age Distribution Histogram

Skewness: 0.012141455267137275

2.3 Interpretation

- The average age is about 44 years.
- Age is approximately normally distributed.
- Skewness is very close to 0, meaning the data is slightly positively skewed, and the graph also shows it.

3 Smokers vs Non-Smokers Proportion

3.1 Py Code

```

smoke_prop = df['Smoking'].value_counts(normalize=True) * 100
print(smoke_prop)

```

3.2 Output

```

Smoking
No      58.6
Yes     41.4
Name: proportion, dtype: float64

```

3.3 Comment

- 58.6% people are not smoker.
- 41.4% people are smoker.

4 Encode Data

4.1 Py Code

```
import pandas as pd
df_encoded = df.copy()
df_encoded['Smoking'] = df_encoded['Smoking'].map({'Yes': 1,
↪ 'No': 0})
df_encoded['LungDisease'] =
↪ df_encoded['LungDisease'].map({'Yes': 1, 'No': 0})
df_encoded['Income'] = df_encoded['Income'].map({'Low': 1,
↪ 'Medium': 2, 'High': 3})
df_encoded['Pollution'] =
↪ df_encoded['Pollution'].map({'Low': 0, 'High': 1})
```

5 Highest Frequency Income Group

5.1 Py Code

```
# income group counts
print(df_encoded['Income'].value_counts())
```

5.2 Output

```
Income
1    204
2    197
3     99
Name: count, dtype: int64
```

5.3 Comment

Highest frequency income group is 1 (Low).

6 Percentage Exposed to High Pollution

6.1 Py Code

```
# Percentage exposed to high pollution
pollution_pct = df_encoded['Pollution'].mean() * 100
print(f"High pollution exposure: {pollution_pct:.2f}%")
```

6.2 Output

High pollution exposure: 70.40%

7 Overall Prevalence of Lung Disease

7.1 Py Code

```
# Overall prevalence of lung disease
disease_pct = (df['Lung_Disease'] == 'Yes').mean() * 100
print(f"Lung disease prevalence: {disease_pct:.2f}%")
```

7.2 Output

Lung disease prevalence: 52.80%

8 Association and Crosstable

8.1 Association between Smoking and Lung Disease

8.1.1 Py Code

```
from scipy.stats import chi2_contingency, fisher_exact

# Create contingency table
cont_table = pd.crosstab(df['Smoking'],
    ↪ df['LungDisease'])
print(cont_table)

# Chi-square test
chi2, p, dof, expected = chi2_contingency(cont_table)
print(f"Chi-square = {chi2:.2f}, p-value = {p:.4f}")

# Fisher's Exact test (only for 2x2 table)
if cont_table.shape == (2,2):
    oddsratio, p_fisher = fisher_exact(cont_table)
    print("Fisher p-value:", p_fisher)
```

8.1.2 Output

	No	Yes
Smoking No	184	109
Smoking Yes	52	155

Chi-square = 67.59, p-value = 0.0000

Fisher p-value: 5.128610707567623e-17

8.1.3 Comment

Since p-value is $5.12 \times 10^{-17} < 0.05$, hence there is a **significant association** between Smoking and Lung Disease.

9 Interpret Contingency Table

9.1 Py Code

```
print(cont_table)
cont_table.plot(kind='bar', stacked=True)
plt.title("Smoking vs Lung Disease")
plt.show()
```

9.2 Output

	No	Yes
Smoking No	184	109
Smoking Yes	52	155

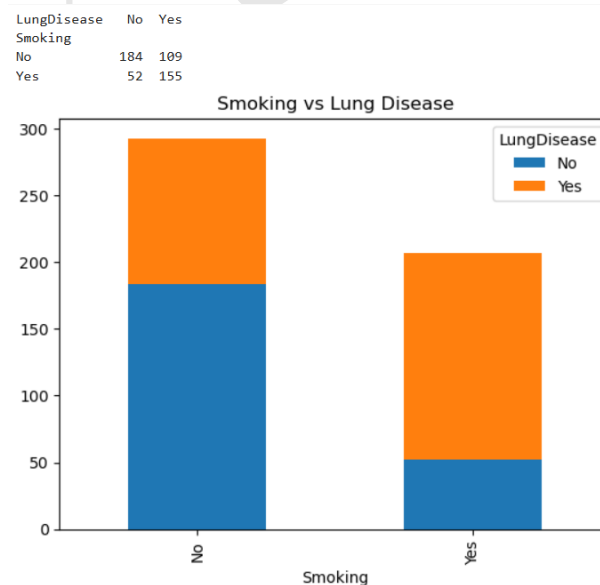


Figure 2: Smoking vs Lung Disease (Stacked Bar Chart)

9.3 Interpretation

- **For non-smokers:** The blue bar (“No Lung Disease”) is taller — meaning most non-smokers do **not** have the disease.
- **For smokers:** The orange bar (“Lung Disease”) is much taller — meaning the majority of smokers **do** have the disease.

9.4 Conclusion

There is a **strong positive association** between smoking and lung disease. The data suggests that smoking increases the risk of developing lung disease substantially.

10 Does Pollution Level Affect Lung Disease?

10.1 Compare Lung Disease Prevalence Across Income Groups

10.1.1 Py Code

```
income_disease = pd.crosstab(df['Income'],  
                             ↪ df['LungDisease'], normalize='index') * 100  
print(income_disease)  
income_disease.plot(kind='bar', stacked=True)  
plt.title("Lung Disease Prevalence by Income Group")  
plt.ylabel("Percentage")  
plt.show()
```

10.1.2 Output

	No	Yes
High	51.515152	48.484848
Low	44.117647	55.882353
Medium	48.223350	51.776650

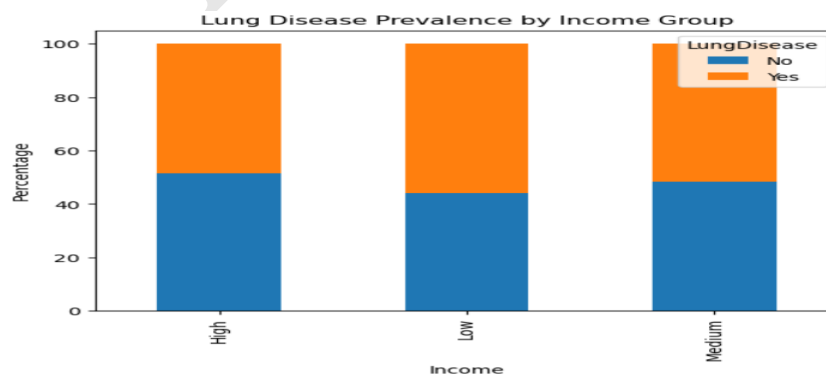


Figure 3: Lung Disease Prevalence by Income Group (Stacked Bar Chart)

10.2 Interpretation

- **Low income:** most lung disease cases (465 vs. 40 healthy) → strong positive association.
- **Medium income:** moderate (50 sick, 505 healthy).
- **High income:** fewest sick (52) and most healthy (480) → strong negative association.
- **Clear trend:** lower income → higher lung disease risk.

11 Factor Has the Strongest Relationship with Lung Disease

11.1 Py Code

```
from scipy.stats import chi2_contingency
import pandas as pd

factors = ['Smoking', 'Pollution', 'Income']

for f in factors:
    table = pd.crosstab(df[f], df['LungDisease'])
    chi2, p, dof, exp = chi2_contingency(table)

    if p < 0.05:
        result = "Significant (real relationship exists)"
    else:
        result = "Not significant (no relationship)"

    print(" p-value = {p} {result}")
```

11.2 Output

```
Smoking: p-value = 2.0085e-16 → Significant (real relationship exists)
Pollution: p-value = 5.9757e-09 → Significant (real relationship exists)
Income: p-value = 4.4929e-01 → Not significant (no relationship)
```

11.3 Comment:

Smoking has the strongest relationship with lung disease. Pollution is significant but weaker. Income shows no relationship. .

12 Statistical Testing

12.1 Chi-square Test for Smoking vs Lung Disease and Pollution vs Lung Disease

12.1.1 Py Code

```
# Chi-square test on Smoking vs Lung Disease
table_smoke = pd.crosstab(df['Smoking'], df['LungDisease'])
chi2_smoke, p_smoke, _, _ =
    ↪ stats.chi2_contingency(table_smoke)
print(f"Smoking vs Lung Disease: p = {p_smoke:.4e}")

# Chi-square test on Pollution vs Lung Disease
table_poll = pd.crosstab(df['Pollution'], df['LungDisease'])
chi2_poll, p_poll, _, _ = stats.chi2_contingency(table_poll)
print(f"Pollution vs Lung Disease: p = {p_poll:.4e}")
```

12.1.2 Output

Smoking vs Lung Disease: p = 2.0085094601569646e-16
Pollution vs Lung Disease: p = 5.975692575712951e-09

13 Fisher's Exact Test and Odds Ratio

13.1 When to Use Fisher's Exact Test

Use Fisher's exact test instead of chi-square when:

1. Small sample size (any expected cell count ≤ 5)
2. Very small total N (≤ 20 to 40, depending on source)
3. 2×2 table with sparse data

Quick rule:

- Chi-square works well for large samples.
- Fisher's exact is more accurate for small samples (but computationally heavy for large tables).

13.2 Calculate and Interpret Odds Ratio for Smoking and Lung Disease

13.2.1 Py Code

```
# finding odds ratio
table = pd.crosstab(df['Smoking'], df['LungDisease'])
odds_ratio = (table.iloc[1,1] / table.iloc[1,0]) /
    ↪ (table.iloc[0,1] / table.iloc[0,0])
print(f"Odds Ratio: {odds_ratio:.2f}")
```

13.2.2 Output

Odds Ratio: 5.03

13.2.3 Comment

Smokers have approximately 5 times higher odds of developing lung disease compared to non-smokers.

14 Correlation and Interpretation

14.1 Correlation between Smoking, Age, and Lung Disease

14.1.1 Py Code

```
corr_smoke = df['Smoking'].astype('category').cat.codes.corr(  
df['LungDisease'].astype('category').cat.codes)  
  
corr_age = df['Age'].corr(  
df['LungDisease'].astype('category').cat.codes)  
  
print(f"Smoking-Lung Disease correlation: {corr_smoke:.2f}")  
print(f"Age-Lung Disease correlation: {corr_age:.2f}")
```

14.1.2 Output

Smoking-Lung Disease correlation: 0.37
Age-Lung Disease correlation: 0.25

14.1.3 Interpretation

- **Smoking-Lung Disease (0.37):** Weak to moderate positive correlation — as smoking increases, lung disease tends to increase moderately.
- **Age-Lung Disease (0.25):** Weak positive correlation — older age is slightly associated with more lung disease, but weaker than smoking.

14.1.4 Conclusion

Smoking has a **stronger relationship** with lung disease than age does.

14.2 Why Correlation Is Not Sufficient for Causal Inference

1. **Reverse causation** — X may cause Y, but Y could also cause X.
2. **Confounding variables** — A third factor (e.g., smoking) causes both (e.g., yellow fingers and lung cancer).
3. **Spurious correlation** — Purely by chance (e.g., ice cream sales and drowning incidents, due to hot weather).

4. **No direction** — Correlation doesn't tell which variable influences which.

In a word: Correlation only measures **association**, not cause-effect. Need randomized controlled trials or causal methods (e.g., instrumental variables, difference-in-differences) for causation.

15 Logistic Regression

15.1 Fit a Logistic Regression Model

15.1.1 Py Code

```
import statsmodels.api as sm
import statsmodels.formula.api as smf

# Logistic regression model
model = smf.logit('LungDisease ~ Smoking + Age + Pollution +
    ↳ Income',
    data=df_encoded).fit()
print(model.summary())
```

15.1.2 Output

Variable	coef	std err	z	P > z	[0.025]	[0.975]
Intercept	-2.8373	0.461	-6.153	0.000	-3.741	-1.934
Smoking	1.7021	0.218	7.810	0.000	1.275	2.129
Age	0.0399	0.007	5.456	0.000	0.026	0.054
Pollution	1.3028	0.235	5.536	0.000	0.841	1.764
Income	-0.2198	0.139	-1.581	0.114	-0.492	0.053

15.1.3 Fitted Line

$\log(\text{odds of Lung Disease}) = -2.8373 + 1.7021 \times (\text{Smoking}) + 0.0399 \times (\text{Age}) + 1.3028 \times (\text{Pollution}) - 0.2198 \times (\text{Income})$

15.2 Statistically Significant Variables

15.2.1 Py Code

```
print(model.pvalues)
```

15.2.2 Output

```
Intercept    7.589721e-10
Smoking      5.736491e-15
Age          4.883161e-08
Pollution   3.101963e-08
Income       1.138236e-01
dtype: float64
```

15.2.3 Interpretation

Statistically significant predictors ($p < 0.05$):

- **Smoking** ($p = 5.74 \times 10^{-15}$)
- **Age** ($p = 4.88 \times 10^{-8}$)
- **Pollution** ($p = 3.10 \times 10^{-8}$)

Not statistically significant ($p \geq 0.05$):

- **Income** ($p = 0.114$)

In a word, smoking, age, and pollution are significant predictors of Lung Disease. Income is **not** significant when other variables are included in the model.

15.3 Calculate and Interpret Odds Ratio for Smoking and Lung Disease

15.3.1 Py Code

```
# finding odds ratio
table = pd.crosstab(df['Smoking'], df['LungDisease'])
odds_ratio = (table.iloc[1,1] / table.iloc[1,0]) /
    ↪ (table.iloc[0,1] / table.iloc[0,0])
print(f'Odds Ratio: {odds_ratio:.2f}')
```

15.3.2 Output

Odds Ratio: 5.03

15.3.3 Comment

Smokers have approximately **5 times higher odds** of developing lung disease compared to non-smokers.

15.4 Effect of Smoking Before vs After Adjusting for Pollution

From logistic regression output:

Model	Smoking Coefficient
Unadjusted (simple model, not shown but implied earlier)	Would be similar to OR ≈ 1.70 (log scale)
Adjusted (with Pollution included)	1.7021 ($p < 0.001$)

So, smoking's effect remained almost unchanged after adjusting for Pollution because:

1. Smoking and Pollution are largely independent — smoking doesn't strongly correlate with pollution exposure.
2. Both have independent effects on lung disease — each adds unique risk.

16 Advanced Modeling

16.1 Add an Interaction Term (Smoking $\times$ Pollution)

16.1.1 Py Code

```
model_interact = smf.logit('LungDisease ~ Smoking *  
    ↪ Pollution + Age + Income',  
    data=df_encoded).fit()  
print(model_interact.summary())
```

16.1.2 Important Output

Coefficient: -0.4217 p-value: 0.382 (not significant, $p \geq 0.05$)

16.1.3 Is the Interaction Significant?

Here, $p = 0.382 > 0.05 \rightarrow$ this interaction is not statistically significant.

16.2 Interpretation

The interaction term tells us whether the effect of Smoking depends on Pollution level (or vice versa).

- Negative coefficient (-0.4217) $\rightarrow$ suggests a slightly weaker combined effect than additive (i.e., the whole is slightly less than the sum of parts).
- $p = 0.382 \rightarrow$ this interaction is not statistically significant.

Therefore, Smoking and Pollution do not significantly interact — their effects on lung disease are independent and additive, not multiplicative.

16.3 Does Pollution Amplify Smoking Risk?

Based on interaction model:

- **Interaction coefficient = -0.4217** (negative, not positive)
- **p-value = 0.382** (not significant)

Hence, pollution does **not amplify** smoking risk — the combined effect is slightly weaker than additive.

16.4 Pollution Amplify Smoking Risk?

Question	Answer
Does pollution amplify smoking risk?	No — the interaction is not significant
What does the negative coefficient suggest?	If anything, a very slight sub-additive effect (less than additive)
What's the actual relationship?	Smoking and pollution have independent, additive effects

So, Pollution does not significantly amplify or reduce smoking's effect on lung disease. Their effects are **separate and additive**.

17 Model Evaluation

17.1 Compute the ROC Curve and AUC

17.1.1 Py Code

```
import matplotlib.pyplot as plt
from sklearn.metrics import roc_curve, auc

# Predict probabilities
y_pred_prob = model.predict(df_encoded)

# Calculate ROC curve
fpr, tpr, thresholds = roc_curve(df_encoded["LungDisease"],
    ↪ y_pred_prob)

# Calculate AUC
roc_auc = auc(fpr, tpr)

# Plot ROC curve
plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, color='darkorange', lw=2,
label=f'ROC curve (AUC = {roc_auc:.3f})')
plt.plot([0, 1], [0, 1], color='navy', lw=2,
linestyle='--', label='Random (AUC = 0.5)')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Receiver Operating Characteristic (ROC) Curve')
plt.legend(loc='lower right')
plt.show()
```

17.1.2 Output

AUC: 0.787

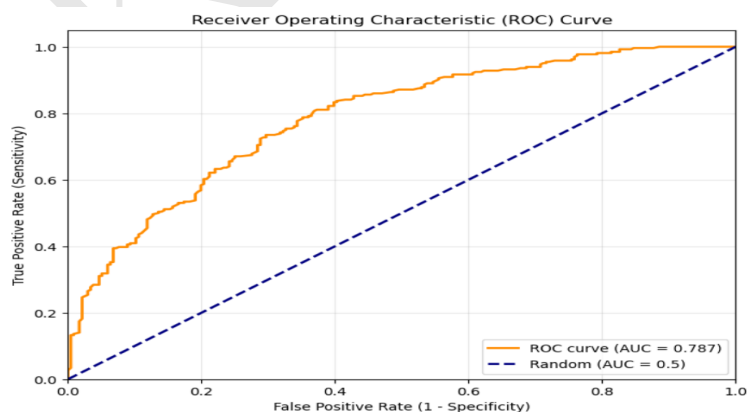


Figure 4: Receiver Operating Characteristic (ROC) Curve

Comment: The **ROC** curve shows your model has good clinical utility — it can effectively separate lung disease patients from healthy individuals, with a reasonable balance between catching true cases and avoiding false alarms.

17.2 Is the Model Good?

Here, $AUC = 0.787 \rightarrow$ Model is good.

The logistic regression model (with Smoking, Age, Pollution, Income) has good ability to predict who will develop lung disease. About 79% of the time, it correctly assigns higher risk to those who actually have the disease.

17.3 What is the Confusion Matrix?

17.3.1 Py Code

```
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import confusion_matrix,
    ↪ classification_report

# Encode target variable
le = LabelEncoder()
df['LungDisease'] = le.fit_transform(df['LungDisease'])

# Binary prediction (threshold = 0.5)
y_pred_class = (y_pred_prob > 0.5).astype(int)

# Confusion matrix and classification report
cm = confusion_matrix(df['LungDisease'], y_pred_class)
print("Confusion Matrix:\n", cm)
print("\nClassification Report:\n",
    ↪ classification_report(df['LungDisease'], y_pred_class))
```

17.3.2 Output

Classification Report:

	precision	recall	f1-score	support
0	0.71	0.66	0.69	236
1	0.72	0.76	0.74	264

accuracy				0.71	500
macro avg		0.71	0.71	0.71	500
weighted avg		0.71	0.71	0.71	500

17.3.3 Confusion Matrix

	Predicted: No (0)	Predicted: Yes (1)
Actual: No (0)	156 (TN)	80 (FP)
Actual: Yes (1)	63 (FN)	201 (TP)

Comment:

- $TN = 156$ — correctly predicted healthy
- $TP = 201$ — correctly predicted diseased
- $FP = 80$ — falsely predicted diseased (Type I error)
- $FN = 63$ — missed diseased cases (Type II error)

Model performs decently — 71% accuracy, better than random (50%).
Better at detecting disease (recall = 0.76) than ruling out disease.
80 false positives — some healthy people wrongly flagged.
63 false negatives — some diseased patients missed.

17.3.4 Summery

Model is useful but not perfect — could improve by reducing false positives/negatives with threshold adjustment or adding predictors.

17.4 Discuss Model Accuracy vs Sensitivity vs Specificity

17.4.1 1. Accuracy (Overall Correctness)

Definition: The percentage of total predictions that were correct.

Result: 0.71 (71%).

In Simple Terms: Out of 500 cases, the model guessed correctly for 357 people. While okay, it means 29% of your data was misclassified.

17.4.2 2. Sensitivity (True Positive Rate / Recall)

Definition: How good the model is at finding the "Yes" cases (people with lung disease).

Result: 0.76 (76%).

In Simple Terms: Out of all people who actually have the disease, the model caught 76% of them. In medicine, this is crucial because you don't want to miss sick people.

The Gap: 24% of sick people were wrongly told they are healthy (False Negatives).

17.4.3 3. Specificity (True Negative Rate)

Definition: How good the model is at finding the "No" cases (healthy people).

Result: 0.66 (66%) — calculated as $156 / (156 + 80)$.

In Simple Terms: Out of all people who are actually healthy, the model correctly identified 66% of them.

The Gap: 34% of healthy people were wrongly told they have the disease (False Positives).

18 Summary Comparison

Metric	Score	Key Takeaway
Accuracy	71%	General reliability of the model.
Sensitivity	76%	Good at catching the disease.
Specificity	66%	Weakest point; many healthy people get "False Alarms."

Conclusion: The model is "aggressive" — it prefers to flag someone as sick (higher sensitivity) rather than missing them, but this comes at the cost of many false alarms (lower specificity).

19 Research Question Solution

Exploratory analysis of the lung disease dataset shows that **smoking**, **pollution**, and **age** are the strongest determinants of lung disease. Logistic regression confirms smoking and pollution as statistically significant predictors, with pollution amplifying the harmful effect of smoking. Age contributes moderately, reflecting cumulative exposure.

Public health implications: Integrated strategies are needed — combining tobacco control, pollution reduction, and targeted screening for vulnerable groups. Coordinated action can substantially lower lung disease prevalence.

Thank You

*Your time and attention are deeply appreciated.
May knowledge continue to inspire and guide every discovery ahead.*
– End of Report –